



NAAC grade 'A'

KCG grade 'A'

ACADEMIC REGULATIONS & SYLLABUS

Faculty of Applied Sciences
P.D. Patel Institute of Applied Science
Department of Physical Sciences
Master of Science Program (Physics)





CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Applied Sciences

ACADEMIC REGULATIONS

Master of Science (Physics) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2022-23
M. Sc. (Physics) Programme

CHARUSAT

FACULTY OF APPLIED SCIENCES ACADEMIC REGULATIONS

Faculty of Applied Sciences

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following are the academic rules and regulations.

1. *System of Education*

The Semester system of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to do a specified course work in the chosen subject of specialization and also complete a project/dissertation if any. Medium of instruction will be English

2. *Duration of Programme*

Postgraduate programme	(M.Sc.)
Minimum	4 semesters (2 academic years)
Maximum	6 semesters (3 academic years)
The maximum limit can be extended by 1 or 2 semester subject to the approval of university on case to case basis.	

3. *Eligibility for admissions*

For the admission to M.Sc., programs in the subject of Biological/Physical/Mathematical/Chemical Sciences a candidate must have obtained a Degree of Bachelor of Science from any recognized University or a Degree recognized as equivalent thereto, with minimum Second Class.

4. *Mode of admissions*

Admission to M.Sc. programme will purely on combined merit of admission test and performance at graduation.

5. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure.

6. Attendance

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be 80%.

7. Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

7.1.2 Final examination will be conducted by the University for 70% of the marks for the course.

7.2 Internal Evaluation

7.2.1 Internal evaluation will be based on internal tests and several other tools of assessment like, quiz, viva, seminar etc., as prescribed by concerned teacher and decided by the faculty.

7.3 Internal Institutional evaluation for practicals

7.3.1 One internal practical test/viva will be conducted per semester totaling to 30 % internal marks for practicals

7.3.2 In “Continuous evaluation” Students shall be evaluated in a continuous manner for their involvement in the practical, aptitude for learning, completion of practical related assignments, regularity in the practicals and record keeping

7.4 University Examination

7.4.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

7.4.2 In order to earn the credit in a course a student has to obtain grade other than FF.

7.5 Performance at Internal & University Examination

- 7.5.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations areas follows

Minimum marks in University Exam per subject	Minimum marks Overall per subject
40%	50%

- 7.5.1. If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grading

- 8.1 The internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Grading Scheme:

Range of Marks (%)	≥80	≥75 <80	≥70 <75	≥65 <70	≥60 <65	≥55 <60	≥50 <55	<50
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

- 8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

- (i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses in the semester
- (ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.
- (iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.

9. Awards of Degree

- 9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:
- 9.1.1 He/ She should have earned at least minimum required credits as prescribed in course structure; and
 - 9.1.2 He/ She should have cleared all internal and external evaluation components in every course; and
 - 9.1.3 He/ She should have secured a minimum CGPA of 5.0 at the end of the programme;
 - 9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies.

10 Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

CGPA	CLASS
7.50 and above	First Class with Distinction
6.00 and Less than 7.50	First class
5.00 and Less than 6.00	Second Class
Less than 5.0	Pass Class

11 Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Faculty of Applied Sciences
P. D. Patel Institute of Applied Sciences
MASTER OF SCIENCE

VISION

To become a pre-eminent national institute imparting science education integrated with research.

MISSION

To engage in education, research, and spread of science for the benefit of society.

PROGRAM EDUCATIONAL OBJECTIVES

The Postgraduate would be able to

PEO-1: Apply the acquired knowledge in various fields of sciences to plan and execute tasks.

PEO-2: Have scientific aptitude and competency in solving local and global problems.

PEO-3 Possess an ethical approach, while executing research projects and scientific writing.

PEO-4 Ability to follow life-long learning.

PEO-5 Understand the importance of team work and possess leadership and entrepreneurship traits.

PROGRAM OUTCOMES

The Post Graduates would be able to

PO1	Knowledge in sciences: Possess knowledge and comprehend the various core and allied courses offered under the opted stream of sciences.
PO2	Planning Abilities: Demonstrate effective planning abilities, including time management, resource management, delegation skills, organizational skills, teamwork, and interpersonal skills.

PO3	Problem analysis: Utilize the principles of scientific enquiry and think analytically, critically with clarity of thought, while solving problems.
PO4	Modern tool usage: Apply appropriate methodologies, methods and procedures, resources to conceptualize the matters.
PO5	Leadership skills: Realize the importance of teamwork and take initiatives to plan and implement a workflow to fulfill tasks with a compassionate attitude.
PO6	Professional Identity: Establish as a professional in the chosen area of the profession with social responsibility.
PO7	Ethics in Science: To understand the value of being ethical and implement its principles while carrying out research and publication, tasks at workplace and in life. To respect and adhere to the rules/regulations/treaties/directives/notifications/laws related to Scientific research and business.
PO8	Communication: Communicate effectively and responsibly with the scientific community and with society at large, such as being able to comprehend and write effective reports, make effective presentations and documentation, and give and receive clear instructions.
PO9	Modern Science and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and help society whenever situations arise.
PO10	Environment and sustainability: Understand the causes of environmental pollution and how it can be reduced and remedied using green technology in industries and environment.
PO11	Life-long learning: Vigilant about changes in the worldwide professional scenario arising out of technological, political, and economic factors. Recognize the need for and undertake steps necessary to fill the gaps recognized to keep oneself professionally competent. Self- assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Department of Physical Sciences

Syllabus

M.Sc. Physics (Sem I-IV)

(Effective from July 2022)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

Subjects for Master of Science (Physics) – 2022 Onwards

Se m.	Subje ct	Subjects	Teaching scheme				Theory Evaluation			Practical Evaluation			Total I +II
			L+T	P	Contact hrs.	Credits	Institute	Univ.	Total-I	Institute	Univ.	Total-II	
I	PS735	Mathematical Methods of Physics	4+1	-	5	5	30	70	100	-	-	-	100
	PS736	Classical Mechanics-1	2+1	-	3	3	30	70	100	-	-	-	100
	PS737	Thermodyna mic and classical Statistics	3+0	3	6	5	30	70	100	15	35	50	150
	PS738	Electronics	3+0	4	7	5	30	70	100	30	70	100	200
	PS739	Computer Programming	1+1	4	6	4	-	-	-	50	100	150	150
	HS70 4E	Academic Speaking	-	2	2	2	-	-	-	30	70	100	100
		TOTAL			29	24			400			400	800

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)

Subjects for Master of Science (Physics) – 2022 Onwards

Se m.	Subjec t	Subjects	Teaching scheme				Theory Evaluation			Practical Evaluation			Total I +II
			L+ T	P	Contact hrs.	Credits	Institute	Univ.	Total-I	Institute	Univ.	Total-II	
II	PS760	Classical Mechanics - 2	3+1	2	6	5	30	70	100	15	35	50	150
	PS761	Electrodyn amics	3+1	2	6	5	30	70	100	15	35	50	150
	PS762	Quantum Mechanics	3+2	2	7	6	30	70	100	15	35	50	150
	PS763	Quantum statistical mechanics	3+1	2	6	5	30	70	100	15	35	50	150
	HS70 5	Academic Writing	-	2	2	2	-	-	-	30	70	100	100
	##-##-##	University Elective	-	2	2	2	-	-	-	30	70	100	100
		TOTAL			29	25			400			400	800

SYLLABI

(Semester – I)

PS735 Mathematical Methods of Physics

1.	Department/Centre/School proposing the course		PHYSICS
2.	Course Title		Mathematical Methods of Physics
3.	L-T-P structure		4-1-0
4.	Credits		5
5.	Course number		PS735
6.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
7.	Pre-requisite(s)		NIL
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Course offered in		Semester -1
11.	Faculty who will teach the course: Dr. Krunal Kachhia/Dr. Shweta Dabhi		
12.	Will the course require any visiting faculty?		No
13.	Course contents: “Complex analysis, vector spaces, operators, inner product, orthogonality and completeness, eigenvalue problem, Hilbert space, Special functions, Fourier transform, FT of Dirac delta functions, Application to the solutions of differential equations, Tensors and Group theory”		
14.	Lecture Outline(with topics and number of lectures)		
	Module no.	Topic	No. of hours
	1	Complex Analysis: Complex variable, Limit of a function of a complex variable, continuity, Differentiability, Analytic functions, Cauchy-Riemann Equations, Harmonic Functions, complex Integration, Cauchy Integral theorem, Cauchy Integral formula, Derivatives of analytic functions, Power series-Taylor’s theorem, Laurent’s theorem, Calculus of Residues, Cauchy’s Residue theorem, Evaluation of real definite integrals	14
	2	Linear Algebra: Vector spaces and operators: Hilbert space, Vector spaces and subspaces, Linear dependence and independence, Basis and Dimensions, linear operators, Inverses. Linear transformation	14

		Matrix algebra: Matrix representation of a linear operator, Change of basis, Eigenvalues and eigenvectors Inner product spaces, Orthogonality: Inner product spaces, Orthogonality, Self adjoint and Unitary transformations, Eigenvalues & eigenvectors of Hermitian & Unitary transformations, Introduction only to GramSchmidt orthogonalization procedure	
	3	Fourier series and integral transforms: Fourier Series : Definition, Dirichlet's condition, Convergence, Fourier Integral and Fourier transform, Convolution theorem, Parseval's identity, Applications to the solution of differential equations, Laplace transform and its properties, Applications to the solution of differential equations, Fourier transform & Laplace transform of Dirac Delta function	14
	4	Special functions: Legendre, Hermite and Laguerre function – Generating function, Recurrence relations and their differential equations, Orthogonality properties, Bessels's function of first kind, Spherical Bessel function, Associated Legendre function, Spherical harmonics.	12
	5	Tensors: symmetric and antisymmetric tensors, covariant and contravariant tensors Group Theory, Preliminaries, Isomorphism and homomorphism, group representation, Lorentz group.	06
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1	Analysis of Complex functions, Analytic functions, Cauchy theorem, Cauchy integral formula, Residue theorem.	04
	2	Vector space, Linear transformations, Eigen values and Eigen vectors	04
	3	Fourier transform, Laplace transform and their applications	03
	4	Legendre function, spherical harmonics	04
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	---	---	---
17.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
	Module 1	Real Functions, Complex Numbers	
	Module 2	Matrices	

18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. F. W. Byron and R. W. Fuller, Mathematics of Classical and Quantum Physics, Vol 1-2, Dover Publications, Inc., New York, 1992. 2. Complex Variables and Applications, J.W. Brown & R.V. Churchill, 7th Ed. 2003, Tata McGraw-Hill. 3. Mary L Boss , Mathematical Methods in Physical Sciences, 3rd Edition, John Wiley & Sons, India. 4. Arfken & Weber, Mathematical Methods for Physicists (6 th edition), Elsevier, Academic Press (2012). 5. Mathematical Methods for Physicists and Engineers, by K.F. Reily, M.P. Hobson and S.J. Bence. 6. Mathematics for Physical Sciences – Mary Boas, John Wiley & Sons. 7. Linear Algebra – Seymour Lipschutz, Schaum Outlines Series- Mc-Graw Hill Edition. 8. Matrices and Tensors in Physics, A. W. Joshi, 3rd Edition, New Age International. 9. Mathematical Methods in Physics – Satyaprakash. 10. Fourier Series - Seymour Lipschutz, Schaum Outlines Series. 11. Fourier Series and Boundary value problems - R. V. Churchill, McGraw Hill.		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	C/C++/Fortran compilers, Codeblock
	19.2	Hardware	Computers
	19.3	Teaching aids (videos, etc.)	Not as such
	19.4	Laboratory	Computer Laboratory
	19.5	Equipment	Computers
	19.6	Classroom infrastructure	Black board and projector
	19.7	Site visits	Not required
	19.8	Others (please specify)	None
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		Nil
	Open-ended problems		Nil
	Project-type activity		Nil
	Open-ended laboratory work		Nil
	Others (please specify)		Nil

Course Outcomes:

In this course the student will

CO1: learn about complex analysis and its applications in Physics

CO2: learn about vectors, vector spaces and Hilbert space and its applications in Quantum Mechanics

CO3: learn about special function and its applications

CO4: learn about Fourier transforms and their solving differential equations arise in Physics

CO5: get introduced to various types of Tensors and learn about group theory and its applications in Physics

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	3	3	3	3	3	3
CO2	3	-	3	3	-	3	3	3	3	3	-
CO3	3	-	3	3	-	-	3	3	3	3	3
CO4	3	-	3	3	-	3	3	3	3	3	3
CO5	3	-	3	3	-	-	3	3	3	3	3

PS736 Classical Mechanics – I

1.	Department/Centre/School proposing the course	PHYSICS	
2.	Course Title	Classical Mechanics – I	
3.	L-T-P structure	2-1-0	
4.	Credits	3	
5.	Course number	PS736	
6.	Course Status (Course Category for Program) PG		
	Programme Core for:	M.Sc. Physics	
7.	Pre-requisite(s)	Concepts of vector space, Newtonian Mechanics	
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Course offered in		Semester -1
11.	Faculty who will teach the course:		Dr. Kinnari Parekh, Dr. Manan Shah
12.	Will the course require any visiting faculty?		No
13.	Course contents: Concepts of Newtonian Mechanics- revisit Central force motions Two body Collisions-scattering theory Generalized coordinates, constraints and Lagrangian equations of motion Hamilton's formalism and equations of motion		
14.	Lecture Outline(with topics and number of lectures)		
	Module no.	Topic	No. of hours
	1	Concepts of Newtonian Mechanics- revisit	5
	2	Central force motions	6
	3	Two body Collisions-scattering theory	4
	4	Generalized coordinates, constraints, D'Alembert's principle and Lagrangian equations of motion	7
	5	Conservation laws and cyclic coordinates. Hamilton's formalism and equations of motion	6
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1	Tutorials and problem solution related to Newtonian Mechanics	1
	2	Tutorials on Central force motions & scattering theory	2

	3	Problem sessions and clarification of doubts related to Generalized coordinates, constraints	2
	4	Tutorials on Lagrangian equations of motion	3
	5	Tutorials on Hamilton's equations of motion	2
	6	Problem sessions and clarification of doubts.	2
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
17.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. <i>Classical Mechanics</i> , H Goldstein, 2nd edition, Addison Wesley Publishing Company. 2. <i>Classical Mechanics</i> – N C Rana and P S Jog, Tata McGraw Hill, 1991 3. <i>Classical Mechanics</i> - Yaswant Waghmare 4. <i>Introduction Classical Mechanics</i> - R. G. Takwale, P. S. Puranik, Tata McGraw Hill 5. <i>Classical Mechanics – System of particles and Hamiltonian Dynamics</i> by Greiner, Springer International Ed. 2006 6. <i>Classical Mechanics</i> by G. Aruldas, PHI LEARNING PVT LTD (2009).		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	Not required
	19.2	Hardware	No hardware required
	19.3	Teaching aids (videos, etc.)	yes
	19.4	Laboratory	Not required
	19.5	Equipment	Not required
	19.6	Classroom infrastructure	No specific facility required
	19.7	Site visits	Not required
	19.8	Others (please specify)	None
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		Nil
	Open-ended problems		Nil
	Project-type activity		Nil

Course Outcomes:

CO1: At the end of the course, student will be able to understand the basic concepts of classical mechanics which includes equation of motion of a particle in 1D, 2D and 3D space, Lagrangian dynamics, Hamiltonian dynamics, central force motion, etc.

CO2: Solve basic problems in classical mechanics such as motion of a system or a particle under various constraints and developing the ability to identify the kinetic energy, potential energy, equation of constraints, etc.

CO3: Student will learn advance classical mechanical theory like variational principle, Hamiltonian dynamics, etc.

CO4: Student will be able to solve 2nd order differential equation by programming in scilab and will be able to analyse the results.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	3	-	3	-	-	3
CO2	3	-	3	-	-	3	-	-	-	-	3
CO3	3	-	3	-	-	3	-	-	-	-	3
CO4	3	-	-	3	3	-	-	3	-	-	3

PS737 Thermodynamic and Classical Statistical Mechanics

1.	Department/Centre/School proposing the course		PHYSICS
2.	Course Title		Thermodynamic and Classical Statistical Mechanics
3.	L-T-P structure		2-1-3
4.	Credits		5
5.	Course number		PS737
6.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
7.	Pre-requisite(s)		NIL
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Course offered in		Semester -1
11.	Faculty who will teach the course:		
12.	Will the course require any visiting faculty?		No
13.	Course contents: Laws of Thermodynamics and Probability, Classical Statistics, Classical Theory of Radiation,		
14.	Lecture Outline (with topics and number of lectures)		
	Module no.	Topic	No. of hours
	1	Fundamentals of Thermodynamics: Laws of thermodynamics, thermodynamics potentials and stability conditions, phase transition	7
	2	Probability: elementary probability theory, distribution functions: Gaussian, Poisson and Maxwell-Boltzmann, central limit theorem, counting principles	7
	3	The Statistical Basis of Thermodynamics : The macroscopic and the microscopic states, Contact between statistics and thermodynamics, The classical ideal gas, The entropy of mixing and the Gibbs paradox, Phase space of a classical system, Liouville’s theorem and its consequences, The microcanonical ensemble, partition function	6
	4	The Canonical Ensemble: Equilibrium between a system and a heat reservoir, A system in the canonical ensemble, Physical significance of the various statistical quantities in the canonical ensemble, partition function, Alternative expressions for the	8

		partition function, The classical systems, Energy fluctuations in the canonical ensemble: correspondence with the microcanonical ensemble, Two theorems — the “equipartition” and the “virial”, A system of harmonic oscillators, The statistics of paramagnetism, Thermodynamics of magnetic systems: negative temperatures,	
	5	The Grand Canonical Ensemble: Equilibrium between a system and a particle-energy reservoir, A system in the grand canonical ensemble, Physical significance of the various statistical quantities, Density and energy fluctuations in the grand canonical ensemble: correspondence with other ensembles, Thermodynamic phase diagrams, Phase equilibrium and the Clausius–Clapeyron equation, Applications of Classical statistical mechanics	8
15.	Brief description of tutorial activities:		
	Module no.	Description	No. of hours
	1	Tutorials on laws of Fundamentals of Thermodynamics	1
	2	Tutorials on Probability	2
	3	Tutorials on The Statistical Basis of Thermodynamics	2
	4	Tutorials on The Canonical Ensemble	2
	5	Tutorials on The Grand Canonical Ensemble	2
16.	Brief description of Practical / Practice activities: performance / Programming language based (e.g.C/C++/Sci-lab/python/matlab/mathematica) practical's		
	Module no.	Description	No. of hours
	Module 1	Plot Newton's Law of Cooling using Differential Equations for different materials.	6
	Module 2	Thermal conductivity of various materials	6
	Module 3	Thermal diffusivity	6
	Module 4	Temperature dependent magnetic susceptibility measurements	6
	Module 5	Temperature dependent dielectric permeability measurements	6
17.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
	Module 1		
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	Text Books: 1. Greiner W., Neise L., Stocker H, Thermodynamics And Statistical Mechanics, Springer 2. B.B. Laud, Fundamentals of Statistical Mechanics, New Age International Publishers 3. R. K. Pathria and Paul D. Beale, Statistical Mechanics, 3 rd Edition, Academic Press, 2012		

	Reference Books: 1. F. Reif, Fundamentals of Statistical and Thermal Physics, International Student Ed., McGraw Hill. 2. Zemansky and Dittman, Heat and Thermodynamics 3. Sheng-Keng Ma, Statistical Mechanics, World Scientific 4. Landau and Lifshitz, Landau theory of phase transition in Statistical Physics		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	C/C++/Sci-lab/python/ matlab/ mathematica
	19.2	Hardware	No hardware required
	19.3	Teaching aids (videos, etc.)	yes
	19.4	Laboratory	Computer Laboratory
	19.5	Equipment	Computers
	19.6	Classroom infrastructure	No specific facility required
	19.7	Site visits	Not required
	19.8	Others (please specify)	None
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		Nil
	Open-ended problems		Nil
	Project-type activity		Nil

Course Outcomes:

CO1: The students should be able to understand, fundamentals of thermodynamics

CO2: The students should be able to understand concepts of classical statistical mechanics

CO3: The students should be able to critically think, analyse and apply problems using statistical mechanical tools to calculate thermodynamics of a given system

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	-	-	-	3
CO2	3	-	3	-	-	-	-	-	-	-	3
CO3	3	3	3	3	-	-	-	-	-	-	-

PS738 Electronics

1.	Department/Centre/School proposing the course		PHYSICS
2.	Course Title		Electronics
3.	L-T-P structure		3-0-4
4.	Credits		5
5.	Course number		PS738
6.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
7.	Pre-requisite(s)		Knowledge of I-V characteristics of transistor and fundamental laws of electronics.
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Course offered in		Semester -1
11.	Faculty who will teach the course:		
12.	Will the course require any visiting faculty?		No
13.	Course contents: Operational amplifier and its applications, timer circuits, combinational and sequential digital circuits		
14.	Lecture Outline(with topics and number of lectures)		
	Module no.	Topic	No. of hours
	1	Operational amplifier (IC-741) and its applications: Basic operational amplifier, Operational amplifier with negative feedback, Voltage series feedback amplifier: close loop voltage gain, input resistance, output resistance, voltage follower. Voltage shunt feedback amplifier: close loop voltage gain, input resistance, output resistance, Inverter. Frequency compensation and slew rate, DC and AC amplifiers. Summing, Scaling and Averaging amplifier, Integrator and Differentiator circuit. Active filters: First order Low pass, High pass, and Band pass filters. Integrator and differentiator, Voltage to current and current to voltage converter, Schmitt trigger	18
	2	Timer (IC-555) & its applications: Internal Structure (Block Diagram) Operation, Astable, Monostable multivibrator and Applications.	5
	3	Combinational digital circuit design:	11

		Standard Gate Assemblies, Arithmetic Functions, Digital Comparator, Parity Checker -Generator, Multiplexer, De-multiplexer, Encoder, Decoder, Digital to Analog Converter, Analog to Digital Converter Tri-State buffer, Read Only Memory (ROM), ROM application	
	4	Sequential digital circuit design: The clocked S-R flip-flop, D flip-flop, J-K flip-flop and T flip-flop, Shift Registers, Counters, Applications of counters	11
15.	Brief description of tutorial activities: - Nil		
	Module no.	Description	No. of hours
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	1	1. closed-loop noninverting op-amp 2. closed-loop inverting amplifier op-amp 3. To design and verify a summing, and scaling amplifier circuit using inverting op-amp (IC741) configuration. 4. To design and verify an averaging amplifier circuit using inverting op-amp (IC741) configuration. 5. To design and verify a differentiator circuit using inverting op-amp (IC741) configuration and plot its output response for sine, square and triangular inputs. 6. To design and verify an integrator circuit using inverting op-amp (IC741) configuration and plot its output response for sine, square and triangular inputs. 7. To design and verify a high pass filter circuit using noninverting op-amp (IC741) configuration and plot its gain versus frequency graph. 8. To design and verify a low pass filter circuit using noninverting op-amp (IC741) configuration and plot its gain versus frequency graph.	32
	2	1. To design and verify astable multivibrator circuit. 2. To design and verify monostable multivibrator circuit	4
	3	1. Study the characteristics of Half-adder and half-subtractor 2. Study the characteristics of full-adder and full-subtractor	8
	4	1. S-R Flip flop and D Flip flop 2. J-K Flip flop and T Flip flop 3. Shift-register 4. Counter	16
17.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
	Module 1	Design aspect of experiments related to Op-Amp (IC-741)	
	Module 2	Design aspect of experiments related to timer IC 555	

	Module 3	Design aspect of experiments related to digital electronics	
	Module 4	Design aspect of experiments related to digital electronics	
18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year		
	1. Ramakant A.Gayakward- Op-Amps and linear integrated circuits- Pearson Education. 2. Electronic Fundamentals and Applications : Integrated and Discrete Systems, John D. Ryder, PHI, Fifth edition. 3. A. Anand Kumar - Fundamentals of Digital Circuits		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	--
	19.2	Hardware	--
	19.3	Teaching aids (videos, etc.)	yes
	19.4	Laboratory	Lab. Equipped to perform electronics experiments
	19.5	Equipment	Power supply, frequency generator, CRO, DMM, assorted items like bread board, wires, electronics components, ICs etc.
	19.6	Classroom infrastructure	No specific facility required
	19.7	Site visits	Not required
	19.8	Others (please specify)	None
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		30-35% of student time for basic circuit design
	Open-ended problems		Nil
	Project-type activity		Nil

Course Outcomes:

At the end of the course, the students will be able

CO1: to understand the fundamentals of working of op-amp and digital electronics

CO2: to explain various applications of analogy and digital electronic based devices

CO3: Design, development and verification of circuit, data measurements, data analysis, handing of laboratory equipments

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3	-	3	-	3	3	-	3
CO2	3	3	3	3	-	3	-	3	3	-	3
CO3	3	3	3	3	3	3	3	3	3	-	-

PS739: Computer Programming

1.	Department/Centre/School proposing the course		PHYSICS
2.	Course Title		Computer Programming
3.	L-T-P structure		1-1-4
4.	Credits		4
5.	Course number		PS739
6.	Course Status (Course Category for Program) PG		
	Programme Core for:		M.Sc. Physics
7.	Pre-requisite(s)		Preferable: Introduction to computer, types of programming languages
8.	Status vis-à-vis other courses		
	8.1 List of courses precluded by taking this course (significant overlap)		
	(a)	Significant Overlap with any UG/PG course of the Dept./Centre/ School	NIL
	(b)	Significant Overlap with any UG/PG course of other Dept./Centre/ School	NIL
9.	Not allowed for		NIL
10.	Course offered in		Semester 1
11.	Faculty who will teach the course: Dr. Manan Shah, Dr. Shweta Dabhi		
12.	Will the course require any visiting faculty?		No
13.	Course contents: Introduction to , C/C++ programming, Control and loop structures, Arrays and functions, Introduction to Python, Elements of computational techniques		
14.	Lecture Outline(with topics and number of lectures)		
	Module no.	Topic	No. of hours
	1	Introduction to language and simple programming: Integer and floating point arithmetic, Precision, Variable types, Arithmetic statements, Input and output statements, Executable and non-executable statements	04
	2	Control and loop structures	02
	3	Arrays and functions: Arrays, Repetitive and logical structures, Subroutines and functions, Operation with files	03
	4	Introduction to Python	02
	5	Elements of computational techniques	04
15.	Brief description of tutorial activities: Problem sessions and clarification of doubts.		
	Module no.	Description	No. of hours
	1	Arithmetic statements, Input and output statements,	02

	2	Control and loop structures	03
	3	Arrays and functions: Arrays, Repetitive and logical structures, Subroutines and functions	03
	4	Programs using Python	03
	5	Elements of computational techniques	03
16.	Brief description of Practical / Practice activities		
	Module no.	Description	No. of hours
	1	Introduction to language and simple programming: Integer and floating point arithmetic, Precision, Variable types, Arithmetic statements, Input and output statements,	8
	2	Control and loop structures: Control structures, Executable and non-executable statements, if...else statements, nested if, switch case, for loop and while loop	8
	3	Arrays and functions: Arrays, Repetitive and logical structures, Subroutines and functions, Operation with files, Creation of executable programs	9
	4	Programs using Python: Libraries to include: NumPy, SciPy, Matplotlib	15
	5	Elements of computational techniques: root of functions, interpolation, extrapolation, integration by trapezoid and Simpson's rule, Solution of first order differential equation using Runge Kutta method.	20
17.	Brief description of module-wise activities pertaining to self-learning component (Include topics that the students would do self-learning from books / resource materials: Do not Include assignments / term papers etc.)		
	Module no.	Description	
	1	Simple program including basic syntax	
	2	Programs including looping structure	
	3	Programs including concepts of arrays and functions	
	4	Programs for plotting and analysing different functions	
18.	Suggested texts and reference materials :		
	1. Programming in C, E Balagurusamy, Tata McGraw-Hill Education 2. Let Us C, Yashwant P Kanetkar, BPB Publications 3. C Language and Numerical Methods by C. Xavier, New Age International Publication 4. Programming with C, Byron Gottfried, Schaum's outlines, McGraw Hill Publications 5. C for Beginners, Madhusudan Mothe, Shroff publishers 6. Computer Oriented Numerical Methods, V. Rajaraman, PHI Learning Publisher 7. Learning Python: powerful object oriented programming, Mark Lutz, O Reilly publications		

	8. Numerical methods in Engineering with Python, by Jaan Kiusalaas, Cambridge University Press 9. Numerical Python: Scientific Computing and Data Science Applications with Numpy, SciPy and Matplotlib, R. Johansson, 2019, Apress. 10. Computational Physics: Problem Solving with Python, 3rd Edition, Rubin H. Landau, 11. Manuel J Páez, Cristian C. Bordeianu, 2015, Wiley. 12. Scientific Computing in Python, Abhijit Kar Gupta, Techno world, 2019. 13. Let Us Python, 2nd Edition, Y. Kanetkar, A. Kanetkar, 2020, BPB Publications.		
19.	Resources required for the course (itemized student access requirements, if any)		
	19.1	Software	“C” or “C++” supported compiler
	19.2	Hardware	Computer system (Linux/Window 10 or higher)
	19.3	Teaching aids (videos, etc.)	Writing board, Projection System
	19.4	Laboratory	Computer lab
	19.5	Equipment	Nil
	19.6	Classroom infrastructure	Yes
	19.7	Site visits	Nil
	19.8	Others (please specify)	
20.	Design content of the course (Percent of student time with examples, if possible)		
	Design-type problems		40% of student time of practice hours
	Open-ended problems		Nil
	Project-type activity		Nil
	Open-ended laboratory work		Nil
	Others (please specify)		Nil

Course Outcomes:

At the end of the course, student would able to

CO1: implement the algorithms and draw flow charts for solving mathematical, physical and engineering problems

CO2: develop ability to design and develop computer programs

CO3: develop confidence for self-education and ability for life-long learning needed for computer languages.

CO4: get familiar with the application of Python programming to describe problems and principles of physics. They will be able to write the source code for simple physical problems.

Course Articulation Matrix											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	-
CO2	-	3	-	3	-	3	-	-	3	-	-
CO3	-	-	-	-	-	3	-	3	3	-	3
CO4	3	-	3	-	-	3	-	3	3	-	3